

tf.compat.v1.sparse_concat Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/sparse_concat

*Concatenates a list of SparseTensor along the specified dimension.
(deprecated arguments)*

Function Signatures

```
tf.compat.v1.sparse_concat( axis, sp_inputs, name=None,  
    expand_nonconcat_dim=False, concat_dim=None,  
    expand_nonconcat_dims=None )  
(concat_dim)
```

Code Examples

```
tf.compat.v1.sparse_concat(  
    axis,  
    sp_inputs,  
    name=None,  
    expand_nonconcat_dim=False,  
    concat_dim=None,  
    expand_nonconcat_dims=None  
)
```

```
tf.compat.v1.sparse_concat(  
axis,  
sp_inputs,  
name=None,  
expand_nonconcat_dim=False,  
concat_dim=None,  
expand_nonconcat_dims=None  
)
```

```
sp_inputs[0]: shape = [2, 3]  
[0, 2]: "a"  
[1, 0]: "b"  
[1, 1]: "c"  
sp_inputs[1]: shape = [2, 4]  
[0, 1]: "d"  
[0, 2]: "e"
```

```
sp_inputs[0]: shape = [2, 3]  
[0, 2]: "a"  
[1, 0]: "b"  
[1, 1]: "c"  
sp_inputs[1]: shape = [2, 4]  
[0, 1]: "d"  
[0, 2]: "e"
```

```
shape = [2, 7]  
[0, 2]: "a"  
[0, 4]: "d"  
[0, 5]: "e"  
[1, 0]: "b"  
[1, 1]: "c"
```

```

shape = [2, 7]
[0, 2]: "a"
[0, 4]: "d"
[0, 5]: "e"
[1, 0]: "b"
[1, 1]: "c"

```

```

[  a] concat [ d e ] = [  a d e ]
[b c ]      [      ]   [b c      ]

```

```

[  a] concat [ d e ] = [  a d e ]
[b c ]      [      ]   [b c      ]

```

Documentation

Concatenates a list of SparseTensor along the specified dimension. (deprecated arguments)

View aliases

Compat aliases for migration

See

Migration guide for
more details.

`tf.compat.v1.sparse.concat`

Concatenation is with respect to the dense versions of each sparse input.
It is assumed that each inputs is a SparseTensor whose elements are ordered

along increasing dimension number.

If `expand_nonconcat_dim` is `False`, all inputs' shapes must match, except for the concat dimension. If `expand_nonconcat_dim` is `True`, then inputs' shapes are allowed to vary among all inputs.

The indices, values, and shapes lists must have the same length.

If `expand_nonconcat_dim` is `False`, then the output shape is identical to the inputs', except along the concat dimension, where it is the sum of the inputs' sizes along that dimension.

If `expand_nonconcat_dim` is `True`, then the output shape along the non-concat dimensions will be expand to be the largest among all inputs, and it is the sum of the inputs sizes along the concat dimension.

The output elements will be resorted to preserve the sort order along increasing dimension number.

This op runs in $O(M \log M)$ time, where M is the total number of non-empty values across all inputs. This is due to the need for an internal sort in order to concatenate efficiently across an arbitrary dimension.

For example, if `axis = 1` and the inputs are

then the output will be

Graphically this is equivalent to doing

Another example, if `'axis = 1'` and the inputs are

if `expand_nonconcat_dim = False`, this will result in an error. But if `expand_nonconcat_dim = True`, this will result in:

Graphically this is equivalent to doing

Returns

Raises